

MH1201 Linear Algebra II

Tutorial 6 (Week 7) – Problems & Solutions

Academic Year 2025/2026, Semester 2
Quantitative Research Society @NTU

August 17, 2026

Overview

This tutorial emphasizes:

- spanning sets and images under surjective linear maps ($V = \text{span}(S) \Rightarrow W = \text{span}(T[S])$);
- RREF pivot criteria for injectivity and surjectivity of linear maps via matrix representations;
- equal-dimension criterion: injective $\Leftrightarrow$ surjective for linear maps between finite-dimensional spaces;
- invertibility under composition and sharp counterexamples for the converse;
- polynomial-space linear operators and existence of polynomial solutions to a (polynomial) ODE constraint;
- constructing explicit isomorphisms between matrix subspaces of equal dimension;
- change-of-basis computation of a matrix representation $[T]_{\gamma}^{\gamma}$.

Contents

Question 1 (Spanning sets and images)	3
Problem	3
Solution	3
Method 1: Write an arbitrary element of W as a linear combination	3
Method 2: Use the identity $T(\text{span}(S)) = \text{span}(T[S])$	4
Method 3: Use the universal property of generated subspaces	5
Method 4: Basis extraction from a spanning set	5
Question 2 (RREF pivots and injective/surjective criteria)	6
Problem	6
Solution	6
Method 1: Reduce to the standard matrix equation $[T(v)]_\beta = [T]_\beta^\alpha [v]_\alpha$	6
Method 2: Rank conditions ($\text{rank}(A)$) and RREF pivot counting	7
Question 3 (Equal dimensions: injective $\Leftrightarrow$ surjective)	8
Problem	8
Solution	8
Method 1: Rank-nullity and dimension inequalities	8
Method 2: Matrix representation (square matrix) and invertibility	8
Question 4 (Invertibility under composition; converse?)	9
Problem	9
Solution	9
Method 1: Verify the inverse by direct composition	9
Method 2: Conceptual structure (what can and cannot be concluded)	10
Question 5 (Polynomial existence for an ODE-type identity)	11
Problem	11
Solution	11
Method 1: Show a linear operator on $\mathcal{P}_n(\mathbb{R})$ is invertible via a triangular matrix	11
Method 2: Trivial kernel + finite-dimensionality implies surjectivity	12
Question 6 (Isomorphism between upper-triangular and symmetric matrices)	13
Problem	13
Solution	13
Method 1: Copy the upper triangle to the lower triangle	13
Method 2: Identify both spaces with $\mathbb{R}^{n(n+1)/2}$ via coordinates	14
Question 7 (Compute $[T]_\gamma^\gamma$ by change of basis)	15
Problem	15
Solution	15
Method 1: Similarity transform $[T]_\gamma^\gamma = P^{-1}AP$	15
Method 2: Compute $T(g_j)$ and solve for γ -coordinates (column-by-column)	16

Question 1 (Spanning sets and images)

Problem

Let

- (i) V and W be vector spaces over the same field,
- (ii) $T \in \mathcal{L}(V, W)$ be surjective onto W , and
- (iii) $S \subseteq V$.

Show that if $V = \text{span}(S)$, then $W = \text{span}(T[S])$.

Remark: Recall that $T[S]$ is the image of S under T , i.e. $T[S] = \{T(v) \in W \mid v \in S\}$.

Solution

Method 1: Write an arbitrary element of W as a linear combination

Assume T is surjective and $V = \text{span}(S)$.

Let $w \in W$ be arbitrary. Since T is surjective onto W , there exists $v \in V$ such that

$$T(v) = w.$$

Since $V = \text{span}(S)$, we can write v as a finite linear combination of elements of S :

$$v = \sum_{i=1}^k a_i s_i, \quad a_i \in \mathbb{F}, \quad s_i \in S.$$

Apply linearity of T :

$$w = T(v) = T\left(\sum_{i=1}^k a_i s_i\right) = \sum_{i=1}^k a_i T(s_i).$$

Each $T(s_i) \in T[S]$, so w is a linear combination of elements of $T[S]$. Hence $w \in \text{span}(T[S])$. Since $w \in W$ was arbitrary, we have shown

$$W \subseteq \text{span}(T[S]).$$

Conversely, $\text{span}(T[S]) \subseteq W$ is automatic because $T[S] \subseteq W$ and $\text{span}(T[S])$ is a subspace of W . Therefore,

$$W = \text{span}(T[S]).$$

$$\boxed{V = \text{span}(S) \text{ and } T \text{ surjective} \implies W = \text{span}(T[S])}.$$

Method 2: Use the identity $T(\text{span}(S)) = \text{span}(T[S])$

We first prove the general identity: for any linear map $T : V \rightarrow W$ and any subset $S \subseteq V$,

$$T(\text{span}(S)) = \text{span}(T[S]).$$

Proof of $\subseteq$: Let $y \in T(\text{span}(S))$. Then $y = T(v)$ for some $v \in \text{span}(S)$, so

$$v = \sum_{i=1}^k a_i s_i \quad (a_i \in \mathbb{F}, s_i \in S).$$

Hence

$$y = T(v) = \sum_{i=1}^k a_i T(s_i) \in \text{span}(T[S]).$$

Thus

$$T(\text{span}(S)) \subseteq \text{span}(T[S]).$$

Proof of $\supseteq$: Let $y \in \text{span}(T[S])$. Then

$$y = \sum_{i=1}^k a_i T(s_i)$$

for some $a_i \in \mathbb{F}$ and $s_i \in S$. By linearity,

$$y = T\left(\sum_{i=1}^k a_i s_i\right).$$

Since $\sum_{i=1}^k a_i s_i \in \text{span}(S)$, it follows that $y \in T(\text{span}(S))$. Hence

$$\text{span}(T[S]) \subseteq T(\text{span}(S)).$$

Therefore,

$$T(\text{span}(S)) = \text{span}(T[S]).$$

Now use the assumptions $V = \text{span}(S)$ and surjectivity of T :

$$W = T(V) = T(\text{span}(S)) = \text{span}(T[S]).$$

Hence

$$\boxed{W = \text{span}(T[S])}.$$

Method 3: Use the universal property of generated subspaces

Since $V = \text{span}(S)$, the set S generates V . Hence every vector of V is obtained from elements of S by taking finite linear combinations.

Because T is linear, it preserves finite linear combinations. Therefore the image $T(V)$ is precisely the subspace generated by the images of elements of S , namely

$$T(V) = \text{span}(T[S]).$$

Since T is surjective, $T(V) = W$. Hence

$$W = \text{span}(T[S]).$$

Therefore,

$$\boxed{W = \text{span}(T[S])}.$$

Method 4: Basis extraction from a spanning set

Since $V = \text{span}(S)$, there exists a basis B of V such that $B \subseteq S$.

Because B is a basis of V , we have

$$V = \text{span}(B).$$

Applying Method 2 to the set B , we get

$$T(V) = T(\text{span}(B)) = \text{span}(T[B]).$$

Since T is surjective,

$$W = T(V) = \text{span}(T[B]).$$

Now $B \subseteq S$, so

$$T[B] \subseteq T[S].$$

Hence

$$\text{span}(T[B]) \subseteq \text{span}(T[S]).$$

Thus

$$W = \text{span}(T[B]) \subseteq \text{span}(T[S]) \subseteq W,$$

because $T[S] \subseteq W$. Therefore,

$$W = \text{span}(T[S]).$$

So again,

$$\boxed{W = \text{span}(T[S])}.$$

Question 2 (RREF pivots and injective/surjective criteria)

Problem

Let

- (i) V and W be finite-dimensional vector spaces over the same field,
- (ii) α and β be ordered bases for V and W , respectively, and
- (iii) $T \in \mathcal{L}(V, W)$.

Establish:

- (1) T is injective if and only if the RREF of $[T]_\beta^\alpha$ has a pivot in every column.
- (2) T is surjective onto W if and only if the RREF of $[T]_\beta^\alpha$ has a pivot in every row.

Solution

Let $m = \dim(V)$ and $n = \dim(W)$. Then $[T]_\beta^\alpha$ is an $n \times m$ matrix over $\mathbb{F}$.

Method 1: Reduce to the standard matrix equation $[T(v)]_\beta = [T]_\beta^\alpha [v]_\alpha$

Let $A := [T]_\beta^\alpha$. By definition of matrix representation,

$$\forall v \in V : \quad [T(v)]_\beta = A [v]_\alpha.$$

The coordinate maps $C_\alpha : V \rightarrow \mathbb{F}^m$ and $C_\beta : W \rightarrow \mathbb{F}^n$ given by $C_\alpha(v) = [v]_\alpha$ and $C_\beta(w) = [w]_\beta$ are linear isomorphisms. Hence T is injective/surjective if and only if the induced map

$$\tilde{T} : \mathbb{F}^m \rightarrow \mathbb{F}^n, \quad \tilde{T}(x) = Ax$$

is injective/surjective.

- (1) T is injective $\iff \tilde{T}$ is injective $\iff \ker(\tilde{T}) = \{0\}$. But $\ker(\tilde{T}) = \{x \in \mathbb{F}^m : Ax = 0\}$, i.e. the solution space of $Ax = 0$. In RREF, having a pivot in *every column* means there are no free variables, hence the only solution to $Ax = 0$ is $x = 0$. Thus $\ker(\tilde{T}) = \{0\}$, i.e. $\tilde{T}$ (and hence T) is injective. Conversely, if some column has no pivot, then there is a free variable and a nonzero solution to $Ax = 0$, so T is not injective. Therefore,

$$T \text{ injective} \iff \text{rref}([T]_\beta^\alpha) \text{ has a pivot in every column.}$$

- (2) T is surjective onto $W \iff \tilde{T}$ is surjective onto $\mathbb{F}^n$. Surjectivity of $\tilde{T}(x) = Ax$ means: for every $y \in \mathbb{F}^n$, the system $Ax = y$ is consistent. In RREF, this holds for all y exactly when there is a pivot in *every row*, i.e. there is no zero row that could create a contradiction $0 = (\text{nonzero})$ after augmentation. Equivalently, pivot in every row means $\text{rank}(A) = n$, so the column space of A equals $\mathbb{F}^n$. Therefore,

$$T \text{ surjective onto } W \iff \text{rref}([T]_\beta^\alpha) \text{ has a pivot in every row.}$$

Method 2: Rank conditions ($\text{rank}(A)$) and RREF pivot counting

Let $A = [T]_{\beta}^{\alpha} \in \mathbb{F}^{n \times m}$. In RREF, the number of pivots equals $\text{rank}(A)$.

- (1) T injective $\iff \text{null}(T) = \{0\} \iff \text{nullity}(T) = 0$. By rank-nullity (in finite dimensions),

$$\dim(V) = \text{rank}(T) + \text{nullity}(T) \implies m = \text{rank}(T).$$

Moreover, $\text{rank}(T) = \text{rank}(A)$ (rank is invariant under change of bases), so $\text{rank}(A) = m$. But $\text{rank}(A) = m$ for an $n \times m$ matrix holds iff there is a pivot in every column (i.e. full column rank). Hence the desired equivalence.

- (2) T surjective onto $W \iff \text{range}(T) = W \iff \text{rank}(T) = \dim(W) = n$. Again $\text{rank}(T) = \text{rank}(A)$, so $\text{rank}(A) = n$, which holds iff there is a pivot in every row (full row rank).

Question 3 (Equal dimensions: injective $\Leftrightarrow$ surjective)

Problem

Let

- (i) V and W be finite-dimensional vector spaces over the same field, and
- (ii) $T \in \mathcal{L}(V, W)$.

Show that if $\dim(V) = \dim(W)$, then T is injective if and only if T is surjective onto W .

Solution

Method 1: Rank–nullity and dimension inequalities

Let $\dim(V) = \dim(W) = n$.

Injective $\Rightarrow$ surjective. If T is injective, then $\text{null}(T) = \{0\}$, so $\text{nullity}(T) = 0$. Rank–nullity gives

$$\dim(V) = \text{rank}(T) + \text{nullity}(T) \Rightarrow n = \text{rank}(T) + 0,$$

so $\text{rank}(T) = n = \dim(W)$. Therefore $\text{range}(T)$ is an n -dimensional subspace of W , hence equals W . Thus T is surjective onto W .

Surjective $\Rightarrow$ injective. If T is surjective, then $\text{range}(T) = W$ so $\text{rank}(T) = \dim(W) = n$. Rank–nullity gives

$$n = \dim(V) = \text{rank}(T) + \text{nullity}(T) = n + \text{nullity}(T),$$

so $\text{nullity}(T) = 0$, i.e. $\text{null}(T) = \{0\}$ and T is injective.

$$\boxed{\dim(V) = \dim(W) \implies (T \text{ injective}) \iff (T \text{ surjective onto } W)}.$$

Method 2: Matrix representation (square matrix) and invertibility

Choose bases α of V and β of W . Then $A = [T]_{\beta}^{\alpha}$ is an $n \times n$ matrix.

- T injective $\iff$ the homogeneous system $Ax = 0$ has only the trivial solution $\iff$ columns of A are linearly independent $\iff \det(A) \neq 0$.
- $\det(A) \neq 0 \iff A$ is invertible $\iff$ for every $y \in \mathbb{F}^n$ there is an x with $Ax = y$, i.e. $\tilde{T}(x) = Ax$ is surjective $\iff T$ is surjective.

The reverse direction is symmetric (surjective for a square matrix implies invertible implies injective).

$$\boxed{A \in \mathbb{F}^{n \times n} \text{ invertible} \iff T \text{ injective} \iff T \text{ surjective.}}$$

Question 4 (Invertibility under composition; converse?)

Problem

Let

- (i) U, V, W be vector spaces over the same field, and
- (ii) $S \in \mathcal{L}(U, V)$ and $T \in \mathcal{L}(V, W)$.
- (a) Suppose that S is invertible from U to V and that T is invertible from V to W . Show that $T \circ S$ is invertible from U to W and that $(T \circ S)^{-1} = S^{-1} \circ T^{-1}$.
- (b) Suppose that $T \circ S$ is invertible from U to W . Is it necessarily true that S is invertible from U to V and that T is invertible from V to W ?

Solution

Method 1: Verify the inverse by direct composition

- (a) Assume $S^{-1} : V \rightarrow U$ and $T^{-1} : W \rightarrow V$ exist, with

$$S^{-1} \circ S = \text{id}_U, \quad S \circ S^{-1} = \text{id}_V, \quad T^{-1} \circ T = \text{id}_V, \quad T \circ T^{-1} = \text{id}_W.$$

Consider $R := S^{-1} \circ T^{-1} : W \rightarrow U$. Then

$$\begin{aligned} (T \circ S) \circ R &= (T \circ S) \circ (S^{-1} \circ T^{-1}) \\ &= T \circ (S \circ S^{-1}) \circ T^{-1} \\ &= T \circ \text{id}_V \circ T^{-1} \\ &= T \circ T^{-1} = \text{id}_W, \\ R \circ (T \circ S) &= (S^{-1} \circ T^{-1}) \circ (T \circ S) \\ &= S^{-1} \circ (T^{-1} \circ T) \circ S \\ &= S^{-1} \circ \text{id}_V \circ S \\ &= S^{-1} \circ S = \text{id}_U. \end{aligned}$$

Thus R is the inverse of $T \circ S$.

$$\boxed{T \circ S \text{ invertible and } (T \circ S)^{-1} = S^{-1} \circ T^{-1}}.$$

- (b) **Not necessarily.** Provide a counterexample.

Take $U = \mathbb{R}$, $V = \mathbb{R}^2$, $W = \mathbb{R}$. Define

$$S : \mathbb{R} \rightarrow \mathbb{R}^2, \quad S(t) = (t, 0), \quad T : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad T(x, y) = x.$$

Then

$$(T \circ S)(t) = T(t, 0) = t,$$

so $T \circ S = \text{id}_{\mathbb{R}}$ is invertible from U to W .

However, S is *not* invertible from $\mathbb{R}$ to $\mathbb{R}^2$ because it is not surjective onto $\mathbb{R}^2$ (e.g. $(0, 1) \notin S(\mathbb{R})$). Also, T is *not* invertible from $\mathbb{R}^2$ to $\mathbb{R}$ because it is not injective (e.g. $T(0, 1) = 0 = T(0, 0)$ with $(0, 1) \neq (0, 0)$).

(b) No: $T \circ S$ invertible does not force S, T invertible.

Method 2: Conceptual structure (what can and cannot be concluded)

- (a) The composition of bijections is a bijection, and the inverse of a composition reverses order. More precisely: since S is bijective $U \rightarrow V$ and T is bijective $V \rightarrow W$, the composition $T \circ S$ is bijective $U \rightarrow W$. Moreover, for all $w \in W$, set $u = S^{-1}(T^{-1}(w))$; then $(T \circ S)(u) = w$, showing $S^{-1} \circ T^{-1}$ is the inverse.
- (b) If $T \circ S$ is invertible, then it is injective and surjective. From injectivity of $T \circ S$ we can deduce S is injective: if $S(u_1) = S(u_2)$ then $(T \circ S)(u_1) = (T \circ S)(u_2)$, hence $u_1 = u_2$. From surjectivity of $T \circ S$ we can deduce T is surjective: for any $w \in W$, there exists $u \in U$ with $T(S(u)) = w$, hence $w \in T(V)$.

However, injectivity of S does not imply surjectivity onto V , and surjectivity of T does not imply injectivity on V . Therefore invertibility of $T \circ S$ does *not* force invertibility of S or T . The counterexample in Method 1 exhibits exactly this phenomenon:

S injective but not surjective, T surjective but not injective, $T \circ S$ invertible.

$T \circ S$ invertible $\Rightarrow S$ injective and T surjective, but not necessarily invertible.

Question 5 (Polynomial existence for an ODE-type identity)

Problem

Let

- (i) $n \in \mathbb{N}_0$, and
- (ii) $Q \in \mathcal{P}_n(\mathbb{R})$.

Show that there exists a $P \in \mathcal{P}_n(\mathbb{R})$ such that

$$Q = (X^2 + X) P'' + 2X P' + P(3).$$

Solution

Method 1: Show a linear operator on $\mathcal{P}_n(\mathbb{R})$ is invertible via a triangular matrix

Define a linear operator $L : \mathcal{P}_n(\mathbb{R}) \rightarrow \mathcal{P}_n(\mathbb{R})$ by

$$L(P) := (X^2 + X)P'' + 2XP' + P(3).$$

(Indeed, $P(3)$ is a constant polynomial, so $L(P)$ is a polynomial; and degrees do not exceed n .)

Step 1: Compute $L(X^k)$ for $0 \leq k \leq n$. For $k = 0$, $P(X) = 1$ gives $P' = 0$, $P'' = 0$, $P(3) = 1$, hence

$$L(1) = 1.$$

For $k \geq 1$, let $P(X) = X^k$. Then

$$P'(X) = kX^{k-1}, \quad P''(X) = k(k-1)X^{k-2}, \quad P(3) = 3^k.$$

Thus

$$\begin{aligned} L(X^k) &= (X^2 + X)k(k-1)X^{k-2} + 2X \cdot kX^{k-1} + 3^k \\ &= k(k-1)(X^k + X^{k-1}) + 2kX^k + 3^k \\ &= \underbrace{k(k+1)}_{\neq 0} X^k + k(k-1)X^{k-1} + 3^k. \end{aligned}$$

Step 2: Triangularity and nonzero diagonal. With respect to the standard ordered basis $(1, X, X^2, \dots, X^n)$, the coordinate vector of $L(X^k)$ involves only degrees $\leq k$ (namely X^k , X^{k-1} , and 1). Hence the matrix of L is *lower triangular*. Its diagonal entry in the X^k position is the coefficient of X^k in $L(X^k)$:

$$\lambda_0 = 1, \quad \lambda_k = k(k+1) \quad (k = 1, 2, \dots, n).$$

Each $\lambda_k \neq 0$ in $\mathbb{R}$, so the lower triangular matrix has nonzero diagonal and is invertible. Therefore L is an isomorphism $\mathcal{P}_n(\mathbb{R}) \rightarrow \mathcal{P}_n(\mathbb{R})$, hence in particular surjective.

So for every $Q \in \mathcal{P}_n(\mathbb{R})$, there exists $P \in \mathcal{P}_n(\mathbb{R})$ such that $L(P) = Q$, i.e.

$$Q = (X^2 + X)P'' + 2XP' + P(3).$$

$$\boxed{\forall Q \in \mathcal{P}_n(\mathbb{R}) \exists P \in \mathcal{P}_n(\mathbb{R}) : Q = (X^2 + X)P'' + 2XP' + P(3).}$$

Method 2: Trivial kernel + finite-dimensionality implies surjectivity

Define L as above. Since $\mathcal{P}_n(\mathbb{R})$ is finite-dimensional, it suffices to show $\ker(L) = \{0\}$, because then

$$\text{nullity}(L) = 0 \implies \text{rank}(L) = \dim(\mathcal{P}_n(\mathbb{R})) \implies \text{range}(L) = \mathcal{P}_n(\mathbb{R}),$$

so L is surjective.

Let $P \in \mathcal{P}_n(\mathbb{R})$ and assume $L(P) = 0$. If $P = 0$ we are done, so suppose $P \neq 0$ and let $\deg(P) = m$ with leading term $P(X) = a_m X^m + \dots$, where $a_m \neq 0$.

Compute the leading term of $L(P)$. We have

$$P'(X) = m a_m X^{m-1} + \dots, \quad P''(X) = m(m-1) a_m X^{m-2} + \dots.$$

Then

$$(X^2 + X)P'' = m(m-1) a_m X^m + (\text{terms of degree} \leq m-1),$$

$$2XP' = 2m a_m X^m + (\text{terms of degree} \leq m-1).$$

Also $P(3)$ is a constant (degree 0), so it does not affect the X^m coefficient. Therefore the coefficient of X^m in $L(P)$ is

$$(m(m-1) + 2m) a_m = m(m+1) a_m.$$

If $m \geq 1$, then $m(m+1) \neq 0$, so $m(m+1) a_m \neq 0$, implying $L(P)$ has degree m and cannot be identically 0. If $m = 0$, then $P(X) = a_0$ and $L(P) = P(3) = a_0$, so $L(P) = 0$ implies $a_0 = 0$.

Thus in all cases, $L(P) = 0$ forces $P = 0$. Hence $\ker(L) = \{0\}$ and L is surjective. Therefore for each Q there exists P with $L(P) = Q$.

$$\boxed{\ker(L) = \{0\} \implies L \text{ is onto } \mathcal{P}_n(\mathbb{R}) \implies \forall Q \exists P.}$$

Question 6 (Isomorphism between upper-triangular and symmetric matrices)

Problem

Let $n \in \mathbb{N}$. Define vector subspaces V and W of $\mathbb{R}^{n,n}$ by

$$V := \{M \in \mathbb{R}^{n,n} \mid M \text{ is upper triangular}\}, \quad W := \{M \in \mathbb{R}^{n,n} \mid M \text{ is symmetric}\}.$$

Construct an isomorphism from V to W (you may rely on past problems).

Solution

We construct an explicit linear bijection $\Phi : V \rightarrow W$ and exhibit its inverse.

Method 1: Copy the upper triangle to the lower triangle

For $U = (u_{ij}) \in V$, define $\Phi(U) = (s_{ij}) \in \mathbb{R}^{n,n}$ by

$$s_{ij} := \begin{cases} u_{ij}, & i \leq j, \\ u_{ji}, & i > j. \end{cases}$$

(i) **Well-defined and lands in W .** By construction $s_{ij} = s_{ji}$ for all i, j , so $\Phi(U)$ is symmetric, i.e. $\Phi(U) \in W$.

(ii) **Linearity.** Let $U, U' \in V$ and $a \in \mathbb{R}$. For each entry (i, j) , the definition of s_{ij} is obtained by selecting either an entry of U or its transpose-entry. Thus entrywise,

$$\Phi(U + U')_{ij} = \Phi(U)_{ij} + \Phi(U')_{ij}, \quad \Phi(aU)_{ij} = a \Phi(U)_{ij},$$

so Φ is linear.

(iii) **Bijectivity via explicit inverse.** Define $\Psi : W \rightarrow V$ by “take the upper-triangular part”: for $S = (s_{ij}) \in W$, set $\Psi(S) = (u_{ij})$ with

$$u_{ij} := \begin{cases} s_{ij}, & i \leq j, \\ 0, & i > j. \end{cases}$$

Then $\Psi(S)$ is upper triangular, so $\Psi(S) \in V$, and Ψ is linear.

Now check compositions:

- For $U \in V$, $(\Psi \circ \Phi)(U)$ retains the upper-triangular entries of $\Phi(U)$, which equal those of U . Hence $\Psi(\Phi(U)) = U$.
- For $S \in W$, since S is symmetric, $\Phi(\Psi(S))$ fills the lower triangle by reflection of the upper triangle, reconstructing S . Hence $\Phi(\Psi(S)) = S$.

Thus $\Psi = \Phi^{-1}$ and Φ is an isomorphism.

$\Phi : V \rightarrow W$ given by reflection of the upper triangle is a linear isomorphism.

Method 2: Identify both spaces with $\mathbb{R}^{n(n+1)/2}$ via coordinates

Both V and W have dimension $\frac{n(n+1)}{2}$:

- V is determined by the entries (i, j) with $i \leq j$ (upper triangle including diagonal): $\frac{n(n+1)}{2}$ free parameters.
- W is also determined uniquely by its upper triangle (since symmetry forces $s_{ji} = s_{ij}$): again $\frac{n(n+1)}{2}$ free parameters.

Define coordinate extraction maps

$$\theta_V : V \rightarrow \mathbb{R}^{n(n+1)/2}, \quad \theta_W : W \rightarrow \mathbb{R}^{n(n+1)/2}$$

that list the upper-triangular entries in row-major order. Both θ_V and θ_W are linear bijections (they are coordinate isomorphisms).

Then

$$\Phi := \theta_W^{-1} \circ \theta_V : V \rightarrow W$$

is a linear bijection, hence an isomorphism. Concretely, this Φ is exactly the “copy upper triangle” map of Method 1.

$$\boxed{V \cong \mathbb{R}^{n(n+1)/2} \cong W \Rightarrow V \cong W.}$$

Question 7 (Compute $[T]_\gamma^\gamma$ by change of basis)

Problem

Define $T \in \mathcal{L}(\mathbb{R}^{3,1})$ by

$$\forall v \in \mathbb{R}^{3,1} : \quad T(v) := \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 3 \\ 0 & -1 & 0 \end{bmatrix} v.$$

Let γ denote the ordered basis

$$\gamma = \left(\begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right) \quad \text{for } \mathbb{R}^{3,1}.$$

Find $[T]_\gamma^\gamma$.

Solution

Let

$$A := \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 3 \\ 0 & -1 & 0 \end{bmatrix}, \quad g_1 = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \quad g_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad g_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

Method 1: Similarity transform $[T]_\gamma^\gamma = P^{-1}AP$

Let P be the change-of-basis matrix from γ -coordinates to standard coordinates; its columns are the basis vectors:

$$P := [g_1 \ g_2 \ g_3] = \begin{bmatrix} -1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then for any coordinate vector $[v]_\gamma \in \mathbb{R}^3$, we have $v = P[v]_\gamma$. Applying T gives

$$T(v) = Av = A(P[v]_\gamma).$$

On the other hand, by definition of $[T]_\gamma^\gamma$,

$$[T(v)]_\gamma = [T]_\gamma^\gamma [v]_\gamma.$$

But also $[T(v)]_\gamma = P^{-1}T(v) = P^{-1}AP[v]_\gamma$. Since this holds for all $[v]_\gamma$, we get

$$[T]_\gamma^\gamma = P^{-1}AP.$$

Compute $P^{-1}AP$ (by direct multiplication), yielding

$$[T]_\gamma^\gamma = \begin{bmatrix} 0 & 2 & 8 \\ -1 & 4 & 6 \\ 0 & -1 & -1 \end{bmatrix}.$$

$$\boxed{[T]_\gamma^\gamma = \begin{bmatrix} 0 & 2 & 8 \\ -1 & 4 & 6 \\ 0 & -1 & -1 \end{bmatrix}}.$$

Method 2: Compute $T(g_j)$ and solve for γ -coordinates (column-by-column)

By definition, the j th column of $[T]_\gamma^\gamma$ is $[T(g_j)]_\gamma$.

Compute images in standard coordinates:

$$T(g_1) = Ag_1 = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 3 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ 0 \end{bmatrix}, \quad T(g_2) = Ag_2 = \begin{bmatrix} 5 \\ 3 \\ -1 \end{bmatrix}, \quad T(g_3) = Ag_3 = \begin{bmatrix} 3 \\ 5 \\ -1 \end{bmatrix}.$$

Now express each $T(g_j)$ as a linear combination of (g_1, g_2, g_3) .

Column 1. Find (a, b, c) such that $ag_1 + bg_2 + cg_3 = T(g_1)$:

$$a \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ 0 \end{bmatrix}.$$

Equating components gives the system

$$\begin{cases} -a + 2b + c = -2, \\ b + c = -1, \\ c = 0. \end{cases}$$

Thus $c = 0$, $b = -1$, and then $-a + 2(-1) + 0 = -2$ gives $a = 0$. Hence

$$[T(g_1)]_\gamma = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}.$$

Column 2. Solve $ag_1 + bg_2 + cg_3 = T(g_2) = (5, 3, -1)^\top$:

$$\begin{cases} -a + 2b + c = 5, \\ b + c = 3, \\ c = -1. \end{cases}$$

So $c = -1$, $b = 4$, and then $-a + 8 - 1 = 5$ gives $a = 2$. Hence

$$[T(g_2)]_\gamma = \begin{bmatrix} 2 \\ 4 \\ -1 \end{bmatrix}.$$

Column 3. Solve $ag_1 + bg_2 + cg_3 = T(g_3) = (3, 5, -1)^\top$:

$$\begin{cases} -a + 2b + c = 3, \\ b + c = 5, \\ c = -1. \end{cases}$$

So $c = -1$, $b = 6$, and then $-a + 12 - 1 = 3$ gives $a = 8$. Hence

$$[T(g_3)]_\gamma = \begin{bmatrix} 8 \\ 6 \\ -1 \end{bmatrix}.$$

Therefore, assembling these columns,

$$[T]_\gamma^\gamma = \begin{bmatrix} 0 & 2 & 8 \\ -1 & 4 & 6 \\ 0 & -1 & -1 \end{bmatrix}.$$

$$\boxed{[T]_\gamma^\gamma = \begin{bmatrix} 0 & 2 & 8 \\ -1 & 4 & 6 \\ 0 & -1 & -1 \end{bmatrix}}.$$